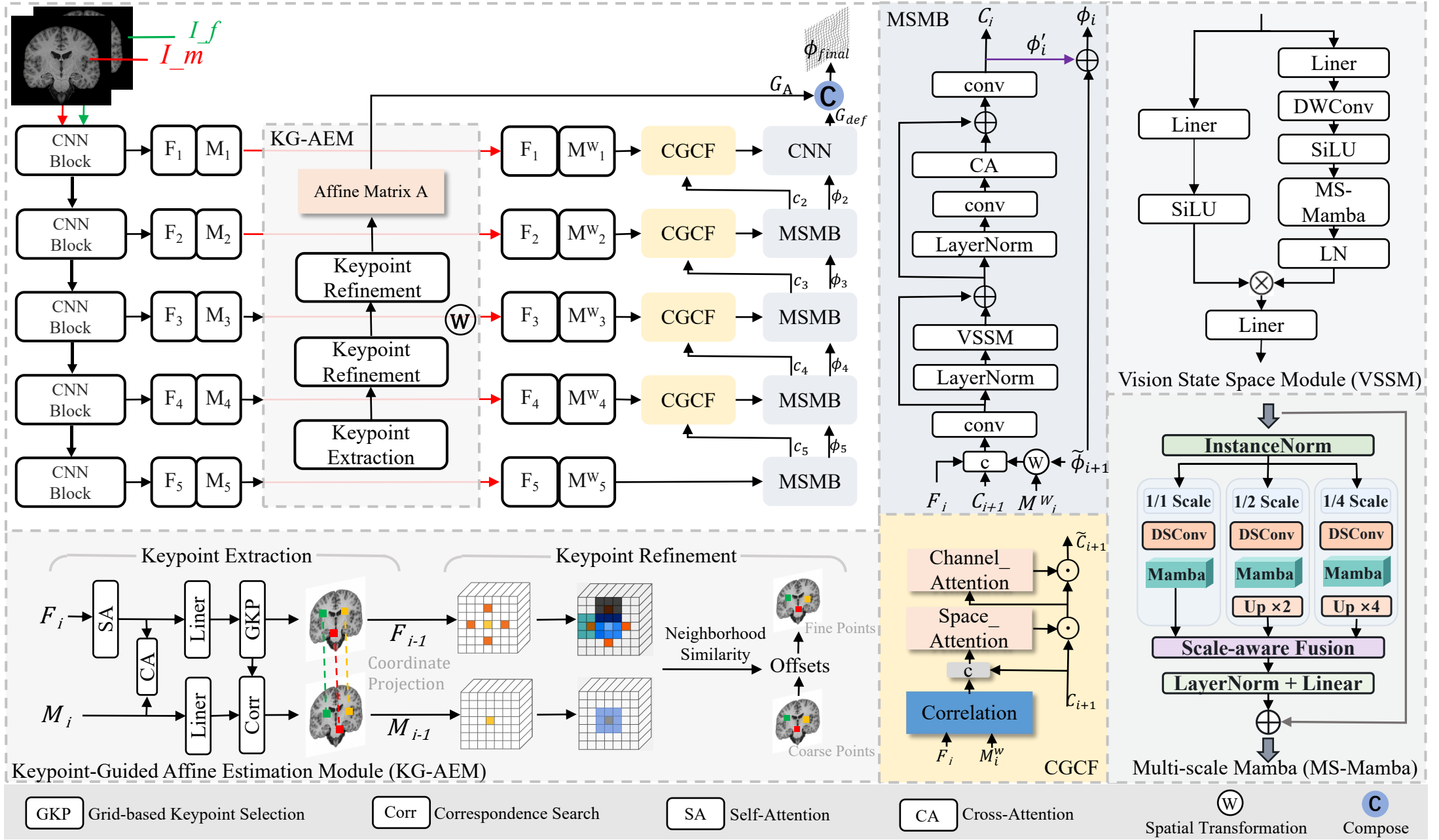

I_f
I_m
CNN Block
F1
M1
KG-AEM
Affine Matrix A
Keypoint Refinement
Keypoint Refinement
Keypoint Extraction
W
MW1
MW2
MW3
MW4
MW5
CGCF
CNN
MSMB
C
GA
Gdef
ϕfinal
ϕ2
ϕ3
ϕ4
ϕ5
c2
c3
c4
c5
F2
M2
F3
M3
F4
M4
F5
M5
MSMB
Ci
ϕi
ϕ'i
conv
CA
conv
LayerNorm
VSSM
LayerNorm
conv
c
W
ϕ̃i+1
Fi
Ci+1
MWi
Liner
DWConv
SiLU
MS-Mamba
LN
Liner
SiLU
Liner
Vision State Space Module (VSSM)
Keypoint Extraction
Keypoint Refinement
Fi
SA
CA
Liner
GKP
Mi
Liner
Corr
Fi-1
Mi-1
Coordinate Projection
Neighborhood Similarity
Offsets
Fine Points
Coarse Points
Keypoint-Guided Affine Estimation Module (KG-AEM)
Channel_ Attention
Space_ Attention
c
Correlation
C̃i+1
Ci+1
Fi
MWi
CGCF
InstanceNorm
1/1 Scale
1/2 Scale
1/4 Scale
DSConv
DSConv
DSConv
Mamba
Mamba
Mamba
Up ×2
Up ×4
Scale-aware Fusion
LayerNorm + Linear
Multi-scale Mamba (MS-Mamba)
GKP Grid-based Keypoint Selection
Corr Correspondence Search
SA Self-Attention
CA Cross-Attention
W Spatial Transformation
C Compose